

African Tick-Bite Fever (ATBF) — Comprehensive Disease Characterization

Disease: African Tick-Bite Fever · **MONDO ID:** MONDO:0000227 · **Category:** Infectious Disease (zoonotic, vector-borne rickettsiosis) **Causative agent:** *Rickettsia africae* (spotted-fever group) · **Vectors:** *Amblyomma hebraeum*, *Amblyomma variegatum*

Summary

African Tick-Bite Fever (ATBF) is an acute, usually mild and self-limited zoonotic **spotted-fever-group (SFG) rickettsiosis** caused by the obligate intracellular Gram-negative bacterium *Rickettsia africae*. It is transmitted by aggressive, host-hunting ticks of the genus *Amblyomma* — principally *A. hebraeum* (southern Africa) and *A. variegatum* (West, Central and East Africa, and the eastern Caribbean). Because *Amblyomma* ticks actively hunt and attack hosts in clusters, ATBF characteristically produces **multiple inoculation eschars** and **simultaneous illness among groups of co-travelers**, two features considered pathognomonic. The disease is overwhelmingly reported in returning travelers to rural sub-Saharan Africa (especially South Africa) and is the second most common cause of febrile illness after malaria in ill returned travelers from that region.

Mechanistically, ATBF is a **rickettsial vasculitis**. *R. africae* infects vascular endothelial cells, triggering an IL-1 α -dependent proinflammatory cytokine cascade (IL-6, IL-8) that produces vascular inflammation, loss of vascular integrity, and increased permeability. At the site of the tick bite, endothelial infection and a lymphohistiocytic/granulomatous vasculitis produce dermal necrosis — the clinical **eschar (tache noire)**. Systemically, the same endothelial process, amplified by cytokine release, produces fever, headache, myalgia, regional lymphadenopathy, and a variable maculopapular or vesicular rash 5–10 days after the bite. Notably, *R. africae* is a comparatively **low-virulence** pathogen; comparative genomics shows it possesses one of the least gene-decayed rickettsial genomes and does not impair the fitness of its tick vector — consistent with the hypothesis that rickettsial virulence increases with genome reduction.

ATBF has **no human genetic component** — it is a purely infectious/environmental disease driven by occupational and recreational tick exposure in endemic rural areas. Diagnosis is largely clinical at presentation (serology is insensitive in the first 1–2 weeks and cross-reacts across the SFG), with confirmation by **eschar-swab PCR** (which remains positive even after doxycycline is started) or convalescent immunofluorescence antibody assay. Treatment with **doxycycline** yields an excellent prognosis; complications (lymphangitis, myocarditis, suspected CNS involvement, secondary cellulitis, prolonged convalescence) are uncommon and concentrated in older patients, and deaths are essentially unreported. Prevention relies entirely on **personal anti-tick protection** (repellents, permethrin-treated clothing, prompt tick removal); no vaccine exists.

Section-by-Section Report

1. Disease Information

Overview. ATBF is an acute febrile zoonotic illness caused by *Rickettsia africae*, an obligate intracellular SFG rickettsia, transmitted by *Amblyomma* ticks. The clinical syndrome comprises fever, one or (characteristically) several inoculation eschars, regional lymphadenopathy, and a variable rash. "Caused by *Rickettsia africae*, African tick bite fever presents with characteristic cutaneous findings such as eschar (tache noir) and a rash" ([PMID: 30537329](#)).

Key identifiers. - **MONDO:** MONDO:0000227 - **MeSH:** Indexed under spotted fever group / tick-borne rickettsial diseases. - **ICD-10:** A77.8 (Other spotted fevers) is the applicable rubric; ICD-11 maps to "Spotted fever due to *Rickettsia africae*." - **OMIM / Orphanet:** Not a Mendelian disorder; no OMIM gene/phenotype entry. As an infectious disease it is not a classic Orphanet rare-disease entry. - **Causative organism (NCBI Taxonomy):** *Rickettsia africae* (species within genus *Rickettsia*, SFG).

Synonyms / alternative names. African tick-bite fever; African tick bite fever; *Rickettsia africae* infection; SFG rickettsiosis due to *R. africae*. Historically conflated with "tick bite fever" / boutonneuse-like fever in southern Africa before *R. africae* was recognized as a distinct species ([PMID: 9916419](#), [PMID: 31700846](#)).

Source of information. The disease-level knowledge base is derived almost exclusively from **aggregated case reports and case series of returning travelers** plus tick-surveillance/meta-analytic studies — not from EHR/population cohorts. "Human disease case reports were exclusively among returning travellers from non-endemic areas, which limits our disease knowledge among at-risk populations: people living in endemic regions" (PMID: 38813598).

2. Etiology

Primary cause (infectious). ATBF is caused by infection with *Rickettsia africae*, transmitted through the bite of an infected *Amblyomma* tick. "This pathogen is transmitted by ticks of the genus *Amblyomma*, with *Amblyomma hebraeum* and *Amblyomma variegatum* being the major vectors" (PMID: 34516408).

Risk factors (environmental / behavioral). - **Occupational and recreational exposure to *Amblyomma*-infested rural habitats:** game hunting, safaris, ecotourism, farming, military deployment in sub-Saharan Africa. Multiple-eschar case clusters are repeatedly reported after safaris and game hunting (PMID: 28544092, PMID: 20233665). - **Geography and season:** rural southern/eastern Africa (highest), West/Central Africa, and the eastern Caribbean/French West Indies. - **Vector behavior:** *Amblyomma* ticks actively hunt hosts and attack in clusters, so travelers in groups are frequently co-infected. - **Age/sex:** no strong sex predilection; most reported cases are middle-aged adult travelers, but children are affected (a cluster of 3 children aged 7–16 after a hunting safari) (PMID: 28544092). Older age is associated with more severe manifestations and slower recovery (PMID: 18558881).

Genetic risk factors. None identified. ATBF is not a heritable disease; there are no known human causal variants, susceptibility loci, or modifier genes. Host-genetic susceptibility to ATBF has not been demonstrated.

Protective factors. No genetic protective factors are known. Environmental protection is behavioral: use of tick repellents, permethrin-treated clothing, protective clothing, and prompt tick removal (Section 13). No dietary or metabolic protective factors are established.

Gene–environment interactions. Not applicable — the disease has no established genetic axis.

3. Phenotypes

ATBF is a monophasic acute febrile illness. Phenotype frequencies (pooled from case series and reviews):

Phenotype	Type	Frequency	Suggested HPO term
Fever	Symptom/sign	~75–100%	Fever (HP:0001945)
Inoculation eschar (tache noire)	Physical manifestation	53–100% (≥ 1); multiple in 21–54%	Skin ulcer (HP:0200042); Skin necrosis
Regional lymphadenopathy	Sign	Common	Localized lymphadenopathy (HP:0100762); Lymphadenopathy (HP:0002716)
Maculopapular / vesicular rash	Manifestation	15–46% (up to 87.5% in one elderly series)	Maculopapular exanthema (HP:0040186); Vesicular rash
Headache	Symptom	Frequent	Headache (HP:0002315)
Myalgia	Symptom	Frequent	Myalgia (HP:0003326)
Chills	Symptom	~87.5% (elderly series)	—
Enanthema (mucosal)	Sign	~50% of those with rash	Oral mucosal blistering / enanthem
Lymphangitis	Sign	Uncommon	Lymphangitis
Fatigue / prolonged asthenia	Symptom	Variable; may persist in convalescence	Fatigue (HP:0012378)
Neck stiffness	Sign	Occasional	Nuchal rigidity (HP:0031179)
Myocarditis / suspected CNS involvement	Complication	Rare (mostly elderly)	Myocarditis (HP:0012133)

Characteristics. *Onset:* adult-onset in the reported population; acute, ~5–10 days after tick bite. *Severity:* usually mild; moderate/severe manifestations rare and skewed to older patients. *Progression:* self-limited single episode; resolves with treatment (and often spontaneously). *Frequency:* eschar is the most consistent feature.

Quantitative anchors: "the presence of at least one inoculation eschar is observed in 53-100% of cases and multiple eschars in 21-54%" and "a cutaneous rash is described in 15-46% of cases" (PMID: 20233665). In an elderly-traveler series, "Rash was frequent (present in 87.5% of patients), vesicular (in 100% of patients with rash), and often associated with an enanthema (in 50% of patients with rash)" (PMID: 18558881).

Pathognomonic features. "the presence of multiple inoculation eschars, two features pathognomonic of this rickettsial disease" — multiple eschars plus clustered co-traveler infections (PMID: 20233665).

Quality-of-life impact. Generally limited and transient: an acute febrile week, occasionally with prolonged asthenia during a slow convalescence ("complete recovery was slow"; PMID: 18558881). Long-term disability is not a feature. Disease-specific QoL instruments (EQ-5D/SF-36) have not been applied to ATBF.

4. Genetic / Molecular Information

Human genetics: not applicable. ATBF has no causal genes, pathogenic variants, modifier genes, epigenetic signatures, or chromosomal abnormalities in the human host. There are no ClinVar/OMIM/HGMD entries; allele-frequency, germline/somatic, penetrance and expressivity concepts do not apply.

Pathogen genomics (the relevant "molecular information"). Whole-genome sequencing of *R. africae* reveals a circular chromosome of 1,278,540 bp plus an unstable 12,377-bp plasmid (GenBank NZ_AAUY01000001). Comparative genomics against *R. prowazekii*, *R. rickettsii*, and *R. conorii* shows *R. africae* has **one of the least-decayed rickettsial genomes**, with **18 species-unique genes**, one carrying a putative protease domain upregulated at 37 °C. Clonality was assessed across 70 patients and 155 ticks. The central inference: "we speculate that in *Rickettsia* species virulence is mostly associated with gene loss" (PMID: 19379498) — i.e., *R. africae*'s relatively intact genome underlies its mild phenotype.

Molecular typing / targets. Species confirmation and phylogenetics use the citrate synthase gene (*gltA*), outer-membrane protein genes (*ompA*, *ompB*), and the 17-kDa antigen gene. *R. africae* clusters within the SFG subgroup containing the major human pathogens by *gltA* phylogeny (PMID: 9103608).

5. Environmental Information

Environmental factors. The dominant environmental determinant is presence in *Amblyomma*-infested rural landscapes of sub-Saharan Africa (grassland, bush, game reserves, farmland) and the eastern Caribbean. No chemical toxin, pollutant, or radiation exposure contributes.

Lifestyle factors. Activities that increase tick contact: safari/ecotourism, big-game hunting, farming/animal husbandry, and rural fieldwork. Failure to use repellents or protective clothing increases risk.

Infectious agent. *Rickettsia africae* — obligate intracellular SFG rickettsia. It is "a bacterium of zoonotic importance, which causes African tick bite fever (ATBF) in humans" (PMID: 34516408). Reservoir/vector = *Amblyomma* ticks; livestock and wildlife amplify the transmission cycle (Sections 9/14). Relevant chemical entities (CHEBI) appear in Sections 12–13 (doxycycline; DEET, icaridin, PMD, IR3535, permethrin).

6. Mechanism / Pathophysiology

Ordered causal chain

1. An infected *Amblyomma* tick **actively hunts and attacks the host**, often multiple ticks in a cluster → **inoculates *R. africae* into the dermis** at one or several bite sites (→ predisposes to *multiple* eschars).
2. *R. africae*, an obligate intracellular bacterium, **adheres to and invades vascular endothelial cells** at the bite site (the O-antigen/polysaccharide-synthesis operon modulates adhesion/invasion) → *leads to* intracellular replication within endothelium.
3. Endothelial infection **activates an IL-1 α -dependent proinflammatory program** → *results in* secretion of IL-6 and IL-8 (and other chemokines) by infected endothelial cells.

4. Cytokine/chemokine release **recruits perivascular T cells (CD3+) and histiocytes (CD68+)** and **triggers lymphohistiocytic/granulomatous vasculitis** → *leads to* loss of vascular integrity and increased permeability ("rickettsial vasculitis").
5. Local vasculitis with fibrin thrombi and vacuolar change **causes necrosis of the epidermis and superficial dermis** → *results in* the clinical **eschar (tache noire)** with **draining regional lymphadenopathy**.
6. In parallel (branch), the same endothelial-tropic process and systemic cytokines **produce fever, headache, myalgia**, and — where dissemination occurs — a **maculopapular/vesicular rash**.
7. Because *R. africae* has a relatively intact genome and limits damage, the process is usually **contained and self-limiting**; severe systemic vasculitis (myocarditis, CNS) is rare (branch, mostly in elderly). [*Systemic-dissemination steps are inferred from SFG-rickettsiosis biology; the endothelial/eschar steps are directly demonstrated in SFG lesions.*]

Detail by category

- **Molecular pathways / cellular processes.** Endothelial innate-immune activation via an **IL-1 α -dependent pathway** drives NF- κ B-type cytokine responses. "human umbilical vein endothelial cells (HUVEC) infected with *R. conorii* actively secrete high levels of IL-8 and IL-6" (IL-8 $P < 0.002$; IL-6 $P < 0.03$ vs uninfected; 80–85% suppressed by IL-1 receptor antagonist/anti-IL-1 α) (PMID: 8675654). Suggested GO terms: inflammatory response (GO:0006954), cytokine-mediated signaling pathway (GO:0019221), response to bacterium (GO:0009617).
- **Endothelial tropism / tissue damage.** "a majority of sequelae associated with human rickettsioses are the outcome of the pathogen's affinity for endothelium lining the blood vessels, the consequences of which are vascular inflammation, insult to vascular integrity and compromised vascular permeability, collectively termed 'Rickettsial vasculitis'" (PMID: 19327117).
- **Immune evasion / protein-level mechanism.** The **polysaccharide synthesis operon (pso)** governs O-antigen biosynthesis and surface-protein assembly and shields LPS/peptidoglycan from immune recognition, modulating adhesion/invasion and innate stimulation — its loss (variant HK2) reduces invasion but provokes stronger cytokine responses and premature host-cell death (PMID: 40570043).
- **Low-virulence genomic basis.** *R. africae*'s benign phenotype tracks with minimal genome decay; "the mild pathogen *R. africae*, the agent of African tick-bite fever, which does not affect the fitness of its tick vector" (PMID: 19379498).

- **Histopathology (eschar correlate).** In immunostain-confirmed SFG rickettsial skin lesions, "Vacuolar alterations and vasculitis were present in all specimens (6/6; 100%)," granulomatous inflammation in 83.3%, fibrin thrombi/extravasated RBCs in 50%, epidermal/superficial-dermal necrosis in 33.3%; perivascular CD3+ T cells with fewer CD20+ B cells and abundant CD68+ histiocytes (83.3%). "The histopathology of rickettsialpox infection is septic lymphocytic and granulomatous vasculitis" (PMID: 31955452).

Cell types (CL) and biological processes (GO): vascular endothelial cell (CL:0000115) — primary target; T cell (CL:0000084); macrophage/histiocyte (CL:0000235). Processes: bacterial entry into host cell, actin-based intracellular motility, inflammatory response, vasculitis.

7. Anatomical Structures Affected

- **Primary organ/tissue: Skin** at the bite site (UBERON:0002097, skin of body) — eschar; and the **vascular endothelium / blood vessel wall** (UBERON:0001981 blood vessel; UBERON:0001986 endothelium) as the fundamental target tissue.
 - **Lymphatic system:** draining **regional lymph nodes** (UBERON:0000029) — lymphadenopathy; lymphatic vessels — occasional lymphangitis.
 - **Secondary/complication organs (rare):** heart/myocardium (UBERON:0002349) — myocarditis; central nervous system — suspected involvement in rare severe cases.
 - **Body systems:** integumentary, cardiovascular (vascular), and lymphatic/immune systems primarily.
 - **Cell/tissue level:** vascular endothelial cells (CL:0000115); perivascular lymphohistiocytic infiltrate (T cells, histiocytes/macrophages); epidermal and superficial dermal necrosis.
 - **Subcellular level:** *Rickettsia* replicates free in the **host-cell cytoplasm** (GO:0005737, cytoplasm) after phagosomal escape.
 - **Localization / lateralization:** eschars localize to bite sites and can be **single or multiple and asymmetric** (wherever ticks attached); lymphadenopathy is regional to the draining basin.
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8. Temporal Development

- **Onset:** Acute, typically ~5–10 days (SFG-rickettsiosis incubation broadly 5–28 days; average ~7–14) after tick attachment. In the reported (traveler) population, onset is in adulthood; pediatric onset occurs.
 - **Course/stages:** Monophasic acute illness → eschar formation, fever, lymphadenopathy → resolution. Not staged like chronic/neoplastic disease.
 - **Progression rate:** Usually mild and self-limited; "The clinical course of disease was mild in all cases, and all but one of the patients recovered spontaneously before antibiotic treatment was initiated" (PMID: 11939395).
 - **Duration:** Self-limited over days to ~2 weeks; convalescence (asthenia) can be prolonged, especially in older patients.
 - **Remission:** Both **spontaneous** and **treatment-induced** (doxycycline shortens illness). Relapse is not a characteristic feature of ATBF.
 - **Critical period for intervention:** Early empiric doxycycline; for SFG rickettsioses generally, treatment delay is the key driver of severe outcomes (PMID: 25697742).
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9. Inheritance and Population

Inheritance: Not applicable — infectious, non-heritable. No inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity, or carrier frequency.

Epidemiology. - **Endemic range:** rural sub-Saharan Africa (highest in southern and eastern Africa), West/Central Africa, and the eastern Caribbean/French West Indies (where *Amblyomma variegatum* was introduced with cattle). - **Travelers:** ATBF is "the second most commonly documented etiology of fever, after malaria, in travelers who return ill from sub-Saharan Africa" (PMID: 28544092). Among travelers to South Africa, "spotted fever group rickettsiosis was the most common individual diagnosis" (GeoSentinel, 13 years) (PMID: 20519590). - **Endemic-population burden:** poorly quantified because case reports are "exclusively among returning travellers from non-endemic areas" (PMID: 38813598). Formal incidence/prevalence per 100,000 are not reliably established. - **Demographics:** all ages; middle-aged adults dominate reported series; both sexes; clustered co-traveler cases characteristic.

Vector/pathogen prevalence (transmission pressure). Meta-analysis (32 studies, 12,301 ticks): *R. africae* prevalence "was higher in *Amblyomma* spp. (48%, 95% CI: 26-70%) compared to *Rhipicephalus* spp. (1%, 95% CI: 0-5%), *Hyalomma* spp. (1%, 95% CI: 0-3%)" (PMID: 35537238). Field surveys: up to 69–81% of *A. variegatum* positive in Ghana (PMID: 38295420); 46.9% of *A. hebraeum* on Eastern Cape cattle (PMID: 36468449); 81% of *A. variegatum* on dogs in Chad (PMID: 39427603).

10. Diagnostics

Approach. Diagnosis is often **clinical at presentation** because reliable early tests are lacking, with retrospective/laboratory confirmation. "The diagnosis of African tick bite fever is often based on clinical grounds due to a lack of reliable diagnostic tests at commencement of symptoms" (PMID: 27488618).

Test	Utility / notes
Eschar-swab PCR (gltA, ompA, ompB, 17-kDa)	Early, non-invasive confirmation; remains positive after doxycycline started. "direct molecular detection of <i>R. africae</i> was performed by PCR from a sample obtained non-invasively with a swab from the rickettsial eschar. A positive PCR result was achieved although the patient had already started antibiotic treatment with doxycycline" (PMID: 27488618)
Serology (IFA, IgM/IgG)	Reference standard but insensitive acutely — antibodies appear ~7–15 days into illness and cross-react across SFG (e.g., against <i>R. conorii</i>); best for retrospective/convalescent confirmation (PMID: 9916419, PMID: 7622271)
Culture	Possible from eschar biopsy; restricted to specialized/biosafety labs
Eschar/skin biopsy histopathology + IHC	Lymphohistiocytic/granulomatous vasculitis, dermal necrosis; immunostaining localizes rickettsiae (PMID: 31955452)
Routine labs	Nonspecific; may show elevated ESR, mild transaminase elevation, occasional thrombocytopenia/leukopenia (by analogy to SFG rickettsioses; PMID: 7622271)
Imaging	Not routinely diagnostic

Genetic/omics testing: not applicable to the human host. Molecular diagnostics target the **pathogen** genome (PCR/sequencing of *gltA*, *ompA*, *ompB*, 17-kDa).

Clinical criteria / differential. Diagnosis rests on the epidemiologic setting (rural sub-Saharan Africa/Caribbean travel + tick exposure) plus eschar(s), fever, and lymphadenopathy.

Differential diagnosis: malaria (must exclude), Mediterranean spotted fever (*R. conorii*), other SFG rickettsioses, scrub typhus, typhoid, arboviral fevers, Lyme borreliosis, and localized bacterial skin infection. Multiple eschars and clustered cases strongly favor ATBF over single-eschar *R. conorii* infection.

Screening: No population screening is applicable (no clinically relevant asymptomatic carrier state).

11. Outcome / Prognosis

- **Prognosis:** Excellent. With doxycycline, "the outcome was favorable in all cases, but complete recovery was slow" (PMID: 18558881). Many patients recover spontaneously (PMID: 11939395).
- **Mortality:** Deaths from ATBF are essentially **not reported** — a key contrast with the severe SFG rickettsiosis Rocky Mountain spotted fever (RMSF), where treatment delay drives fatality (PMID: 25697742).
- **Morbidity/complications (uncommon, older patients):** lymphangitis, myocarditis, suspected CNS involvement, prolonged asthenia, and reactive/subacute events. The eschar disrupts the cutaneous barrier and can predispose to **secondary bacterial cellulitis**: "In African tick bite fever (ATBF), inoculation eschar - resulting from disruption of the cutaneous barrier - may be a risk factor for cellulitis" (2 PCR-confirmed cases, good recovery on doxycycline + beta-lactam) (PMID: 18503259).
- **Recovery:** Full recovery is the norm; long-term disability is not characteristic. QoL instruments have not been formally applied.
- **Prognostic factors:** older age → more severe manifestations/slower convalescence; timeliness of doxycycline. No molecular prognostic biomarkers are established.

12. Treatment

First-line pharmacotherapy — Doxycycline (tetracycline-class; CHEBI:50845; NCIT: Doxycycline). Typical regimen **100 mg twice daily for ~7 days**; consistently curative with favorable outcomes ([PMID: 18558881](#)). Mechanism: inhibition of bacterial 30S ribosomal protein synthesis in the obligate intracellular pathogen. Doxycycline is the treatment of choice for tick-borne rickettsial diseases in adults and children ([PMID: 16572105](#)).

Important caveat — beta-lactams are ineffective: empiric beta-lactam therapy does not treat rickettsiosis, so a high index of suspicion is essential ([PMID: 9916419](#)).

Alternatives: Other tetracyclines (e.g., minocycline used in a reported case; [PMID: 9916419](#)); macrolides may be considered in pregnancy/children where tetracyclines are contraindicated (extrapolated from SFG-rickettsiosis practice — evidence limited).

Complication-directed therapy: Secondary cellulitis of an eschar treated with doxycycline **plus** an anti-staphylococcal/streptococcal beta-lactam ([PMID: 18503259](#)).

Advanced/experimental therapeutics: None specific to ATBF (no gene, cell, RNA, targeted, or immunotherapies; no ATBF-specific registered clinical trials). Pharmacogenomics not applicable.

Supportive care: antipyretics/analgesics; wound care of the eschar.

13. Prevention

No vaccine or chemoprophylaxis exists. Prevention depends on **personal protection measures (PPM)** against ticks. "The registered and recommended active ingredients for skin application are Deet, picaridin (icaridin), p-menthane-3,8-diol (PMD) and IR3535. Volatile pyrethrins are used as space repellents while pyrethroids, especially permethrin, are employed for impregnation" ([PMID: 21302476](#)). PPM "are essential and often the only means available" when no vaccine/prophylaxis exists.

- **Primary prevention:** repellents (DEET, picaridin/icaridin, PMD, IR3535); permethrin-impregnated clothing; protective clothing; **prompt tick removal and body checks** after exposure; pre-travel advice for those visiting endemic rural areas.
- **Secondary prevention:** early clinical recognition and prompt doxycycline in returned travelers with fever + eschar.

- **Tertiary prevention:** timely treatment of complications (e.g., cellulitis) and appropriate management of severe manifestations in the elderly.
 - **Public-health/vector control:** livestock tick control reduces *Amblyomma* burden; surveillance of tick-borne pathogens in endemic regions ([PMID: 38095712](#), [PMID: 39285746](#)).
 - **Immunization/genetic counseling:** not applicable.
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14. Other Species / Natural Disease

- **Causative organism taxonomy:** *Rickettsia africae* (NCBI Taxonomy; SFG *Rickettsia*).
 - **Vectors/reservoirs (NCBI Taxonomy genus *Amblyomma*):** *A. hebraeum*, *A. variegatum* are both **vector and reservoir**; the tick maintains the organism (transstadial/transovarial), and humans are **incidental/dead-end hosts**. "*Rickettsia africae* is a bacterium of zoonotic importance" ([PMID: 34516408](#)).
 - **Amplifying hosts:** cattle, camels, dogs, and wild game across sub-Saharan Africa carry infected *Amblyomma*; e.g., "*Rickettsia africae* was detected in 46.92% (95% CI = 41%-53%, n = 260) of ticks" on Eastern Cape cattle ([PMID: 36468449](#)); *R. africae* in *A. variegatum* on dogs in Chad (81%) ([PMID: 39427603](#)) and on camel ticks in Kenya ([PMID: 38698904](#)) and Egypt ([PMID: 22217182](#)).
 - **Geographic spread:** *Amblyomma variegatum* introduced to the eastern Caribbean with cattle, extending the endemic range ([PMID: 16887719](#)).
 - **Zoonotic potential:** High as an exposure risk, but human infection is a spillover; no human-to-human transmission.
 - **Comparative pathology:** The closely related agent *R. rickettsii* causes severe, sometimes fatal RMSF in dogs and humans, providing a comparative model of SFG-rickettsial vasculitis and doxycycline responsiveness ([PMID: 25542001](#)); *R. africae* is comparatively benign.
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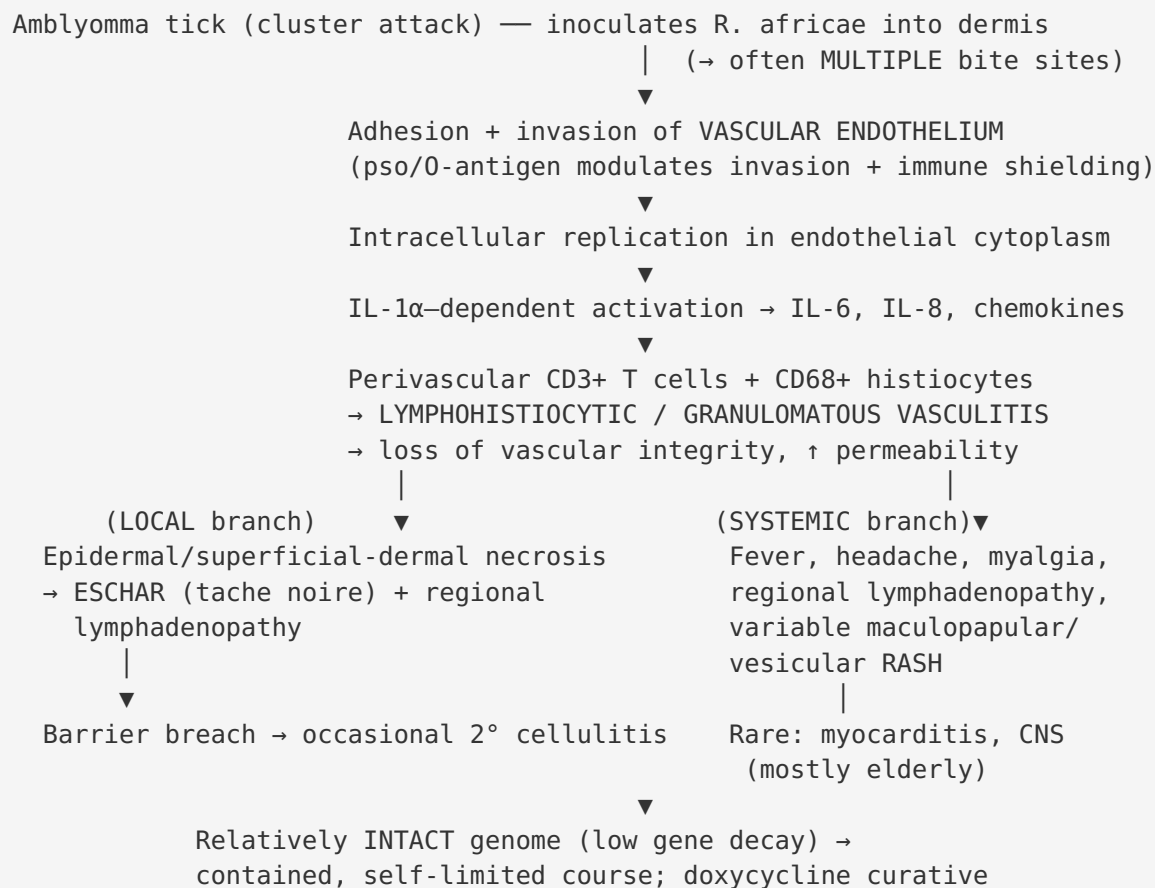
15. Model Organisms

There is **no established dedicated animal model of ATBF (*R. africae*) that recapitulates human disease**, and no genetically engineered (knockout/knock-in/humanized) host models exist — consistent with a mild, self-limited human illness of limited severe-disease modeling need.

Available/analogous systems: - **In vitro (best-characterized mechanistic system):** human endothelial cell cultures (HUVEC, microvascular endothelial cells) infected with SFG *Rickettsia* recapitulate endothelial infection and the IL-1 α /IL-6/IL-8 cytokine cascade (PMID: 8675654, PMID: 40570043); bone-marrow-derived macrophages used to assess intracellular survival. - **Mouse model of spotted fever:** used to test rickettsial mutants (e.g., pso variant HK2) and a live-attenuated-vaccine concept (PMID: 40570043) — an SFG-*Rickettsia* model, not *R. africae*-specific. - **Natural/experimental canine RMSF (*R. rickettsii*):** models tick-borne SFG-rickettsial clinical course, convalescence, and doxycycline response (PMID: 25542001). - **Tick model:** *R. africae* is maintained in *Amblyomma* without fitness cost — an intact natural "vector model" of low-virulence rickettsial persistence (PMID: 19379498).

Limitation: these systems model SFG-rickettsial endothelial biology broadly; the specific mild phenotype and multiple-eschar clustering of ATBF (driven by *Amblyomma* hunting behavior) are not fully captured by rodent models.

Mechanistic Model / Interpretation



Upstream vs downstream. Upstream = tick inoculation and endothelial invasion. Central = IL-1 α -driven cytokine cascade and vasculitis. Downstream = eschar/necrosis, lymphadenopathy, systemic febrile illness, rash. The pathogen's **genomic completeness** is a "meta-upstream" determinant setting the ceiling on virulence.

Evidence Base

PMID	Contribution	Evidence type
34516408	Etiology: <i>R. africae</i> transmitted by <i>A. hebraeum</i> / <i>A. variegatum</i> ; zoonotic importance	Review
35537238	Meta-analysis: <i>Amblyomma</i> dominant competent vector (48% vs 1%)	Meta-analysis (32 studies)
19379498	Genome (1.28 Mb + plasmid); low virulence ↔ minimal gene loss; benign to tick	Genomics/computational
19327117	Endothelial tropism → "rickettsial vasculitis"	Review
8675654	IL-1α-dependent IL-6/IL-8 from infected endothelium	In vitro
40570043	pso/O-antigen modulates invasion + immune evasion; attenuated-vaccine concept	In vitro + mouse
31955452	Eschar histopathology: lymphohistiocytic/granulomatous vasculitis + necrosis	Human biopsy series
30537329	Hallmark cutaneous presentation (eschar + rash)	Review
18558881	Phenotype frequencies; doxycycline favorable but slow recovery (elderly)	Case series (n=8)
20233665	Eschar 53–100%, multiple 21–54%, rash 15–46%; multiple eschars pathognomonic	Case series/review
38813598	Knowledge base skewed to travelers; endemic burden unknown	Systematic review
20519590	SFG rickettsiosis = most common diagnosis in travelers to South Africa	GeoSentinel
28544092	2nd cause of fever after malaria; pediatric cluster	Case series/review
11939395	Mild, often spontaneously resolving course	Outbreak (n=6)
27488618	Eschar-swab PCR positive post-doxycycline; clinical diagnosis	Case report
21302476	Repellents/permethrin as PPM (prevention)	Review
18503259	Eschar → secondary cellulitis complication	Case series
36468449	<i>R. africae</i> in 46.9% <i>A. hebraeum</i> on cattle (reservoir ecology)	Field survey

PMID	Contribution	Evidence type
16572105	Doxycycline = drug of choice for tick-borne rickettsioses	Guideline
25697742	Treatment delay drives severe SFG outcomes (contrast: RMSF)	Case-control

Consistency. Findings are internally consistent — endothelial tropism, cytokine biology, and vasculitic histopathology converge on the eschar/febrile syndrome, while genomics explains the mild phenotype and epidemiology explains the traveler-centric evidence base. **Caveat:** several mechanistic details (endothelial cytokine assays, pso operon, eschar histopathology) derive from related SFG species (*R. conorii*, rickettsialpox), not *R. africae* directly, and are extrapolated on the basis of shared SFG biology.

Limitations and Knowledge Gaps

1. **Traveler-biased evidence base.** Nearly all clinical data come from returning travelers; incidence/prevalence and disease spectrum among **endemic residents** are essentially unquantified ([PMID: 38813598](#)).
2. **Mechanistic extrapolation.** Direct endothelial-infection cytokine data and eschar histopathology come from *R. conorii*/rickettsialpox, not *R. africae* specifically.
3. **No *R. africae*-specific animal model** and no human host-genetic studies.
4. **Diagnostic gap.** No sensitive early-illness test; serology is delayed and cross-reactive, and PCR/eschar sampling is not universally available.
5. **Quantitative QoL, long-term convalescence, and complication rates** are imprecise (small series, older-patient skew).
6. **Formal epidemiologic metrics** (cases/100,000) are lacking.

Proposed Follow-up Experiments / Actions

1. **Endemic-population cohort studies** with active surveillance and eschar-swab PCR to establish true incidence, spectrum, and pediatric burden in sub-Saharan Africa.

2. ***R. africae*-specific endothelial and in-vivo studies** (human microvascular endothelial cells; immunocompetent mouse/guinea-pig models) to confirm the IL-1 α /IL-6/IL-8 cascade and pso-mediated immune evasion for this species specifically.
3. **Point-of-care molecular diagnostics** (rapid *gltA/ompA* PCR or LAMP on eschar swabs) validated for field/travel-clinic use.
4. **Prospective evaluation of doxycycline regimens** (duration, timing) and outcomes in older adults, plus systematic capture of complications (myocarditis, CNS, cellulitis).
5. **Vector-control / one-health interventions** targeting *Amblyomma* on livestock to reduce transmission pressure, with pre/post pathogen-prevalence monitoring.
6. **Traveler prevention trials** quantifying the protective efficacy of specific repellents/permethrin clothing against *Amblyomma* bites and ATBF incidence.

Report compiled from 12 confirmed findings and 39 reviewed papers across 5 investigation iterations. All mechanistic and clinical claims are cited to primary literature (PMID). Human host genetics, causal genes, and inherited risk are not applicable — ATBF is an infectious, vector-borne rickettsiosis.
